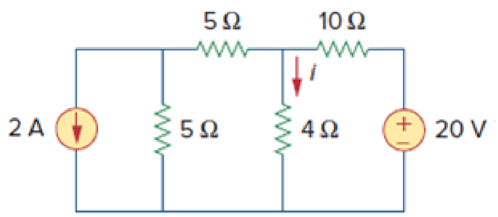
Chapter 4, Problem 22P

Problem

For the circuit in Fig. 4.90, use source transformation to find *i*.

Figure. 4.90:



Step-by-step solution

Step 1 of 4

We transform the two sources to get the circuit as shown in Fig-(a).

Step 2 of 4

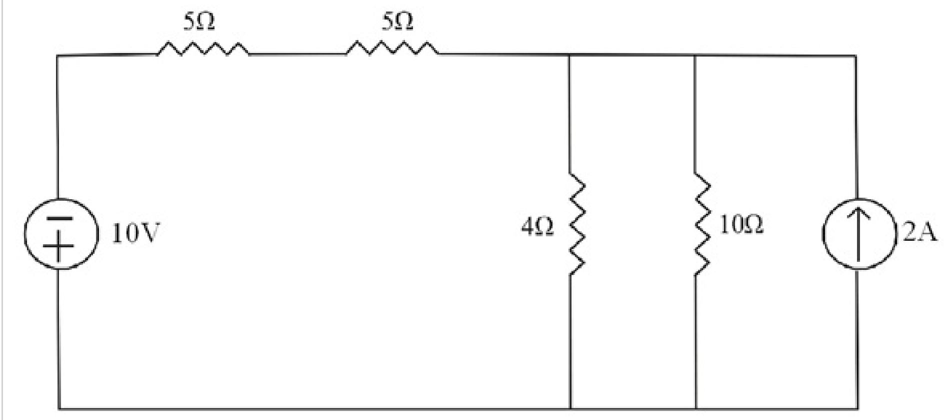


Fig-(a)

Step 3 of 4

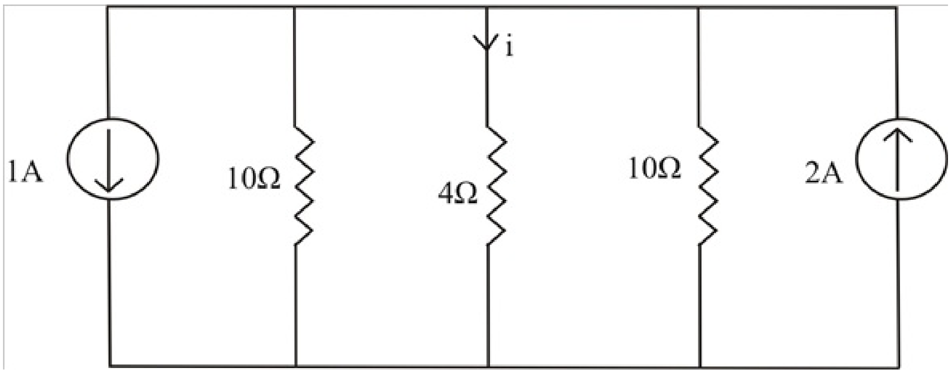


Fig-(b)

We now transform only the voltage source to obtain the circuit in Fig-(b).

$$10\Omega \,||\, 10\Omega = \frac{10\times 10}{10+10}$$

$$10\Omega \,||\, 10\Omega = \frac{100}{20}$$

$$10\Omega \,||\, 10\Omega = 5\Omega$$

$$i_1 = \left[\frac{5}{5+4}\right]\times(2-1)$$

$$i_1 = \frac{5}{9}\times 1$$

$$\boxed{i_1 = 555.5\,\text{mA}}.$$